

KARL HILL

STAFF AEROSPACE SOFTWARE ENGINEER

CROSS-PROGRAM TECHNICAL LEADERSHIP | MISSION SOFTWARE & PLATFORM ENGINEERING

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SUMMARY

Staff Aerospace Software Engineer and technical leader with 25+ years delivering mission-critical software across national security, aerospace, NASA, and enterprise environments. Leads cross-program engineering across multiple teams, shared repositories, partner organizations, and deployment environments while remaining hands-on in Python, distributed systems, integration, and developer tooling. Builds the engineering systems - architecture, CI/CD, DevSecOps, testing, release governance, and standards - that enable teams to deliver reliable software at scale.

LEADERSHIP & ENGINEERING SCOPE

Cross-Program Technical Leadership | Software Architecture | Platform Engineering & Developer Experience | Distributed Systems & Integration | DevSecOps & CI/CD | Engineer Development | Agile Delivery

PROFESSIONAL EXPERIENCE

Staff Aerospace Software Engineer

SEPT 2025 – PRESENT

Jacobs National Security · Chantilly, VA

- Provide cross-program technical leadership across multiple aerospace mission-software efforts, aligning internal and partner teams on engineering standards, shared interfaces, integration strategy, and release readiness.
- Lead technical execution for a core team of about 10 engineers across roughly 20 Python repositories and multiple deployment environments; sequence work, resolve cross-team dependencies, and drive integration and delivery.
- Design and develop mission software, distributed integrations, shared contracts, messaging capabilities, and service orchestration spanning RabbitMQ and ActiveMQ.
- Build and evolve platform engineering and DevSecOps capabilities including Bitbucket Pipelines, automated testing, dependency management, repository standards, quality/security gates, and release automation.
- Provide technical direction through architecture and design reviews, code review, interface decisions, repository governance, and resolution of cross-team implementation and integration issues.
- Onboarded and coached approximately six engineers through technical feedback, development guidance, engineering standards, and structured growth plans while leading Agile execution across team boundaries.

Lead Software Engineer

DEC 2017 – SEPT 2025

SSAI / NASA Goddard Space Flight Center · Greenbelt, MD

- Architected and developed an AWS-based platform generating near-real-time flood and surface-water products from satellite and geospatial data; co-authored the peer-reviewed GeoHorizons publication describing the system.
- Helped rebuild NASA Earth Observatory, serving approximately 1.5 million monthly visitors; the team received a NASA Group Achievement Award.
- Delivered LAADS DAAC Find Data search, ordering, and near-real-time access, with GitLab CI/CD and Kubernetes-based web delivery alongside existing archive services.
- Led modernization of legacy scientific processing workflows into containerized services supported by automated CI/CD and Kubernetes deployment.
- Built an automated content-registry workflow that improved scientific data-collection efficiency by approximately 60%, and developed Ceph-based file and metadata services for large datasets.
- Led Agile technical delivery across engineers, scientists, operations teams, and program stakeholders while strengthening testing, code review, documentation, and production-readiness practices.

DETAILS

Washington, DC

(202) 599-1442

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LINKS

LinkedIn

linkedin.com/in/khill

GitHub

github.com/karlhillx

Website

karlhill.com

AREAS OF EXPERTISE

- Cross-Program Technical Leadership
- Software Architecture
- Platform Engineering & Developer Experience
- Distributed Systems & Integration
- DevSecOps & CI/CD
- Engineer Development
- Agile Delivery

PROFESSIONAL EXPERIENCE

Senior Software Engineer

JAN 2016 – DEC 2017

InformedDNA · Washington, DC

- Designed and delivered a Laravel case-management platform connecting counseling workflows, documentation, and billing.
- Built CRM enhancements for customer lifecycle workflows and reporting, and tightened maintenance and security processes with operations.

Senior Software Engineer

JUN 2012 – MAR 2015

Ticomix, Inc. · Washington, DC

- Delivered SugarCRM solutions for more than 20 clients, including the Virginia Department of Transportation and Kastle Systems.
- Improved backlog management and delivery practices across client projects.

Earlier Engineering Experience

1997 – 2012

Sabre Corporation · Dante Inc. · Visitar Inc. · Verizon Business

- Held software engineering and principal-level roles across travel, enterprise CRM, telecommunications, and managed security.
- At Sabre, built new PHP applications and added features to existing ones for large-scale travel systems.

AREAS OF EXPERTISE

Cross-Program Technical Leadership · Software Architecture · Platform Engineering & Developer Experience · Distributed Systems & Integration · DevSecOps & CI/CD · Engineer Development · Agile Delivery

PUBLICATIONS

Peer-reviewed research: A web-based high-resolution global water and flood mapping platform. Co-author, A web-based high-resolution global water and flood mapping platform, GeoHorizons (AGU / Geological Society of London), 2026. Software engineering for a NASA-supported system delivering near-real-time, high-resolution satellite-derived water and flood products worldwide. Author contribution (CRediT): Software (Equal); Writing – review & editing (Equal). doi:10.1144/gh2025-7

EDUCATION

Computer Science coursework, University of Maryland, Baltimore County

Associate of Arts, General Studies – Computer Science Curriculum, Howard Community College

Project Management Certificate, Rutgers University

CERTIFICATIONS

SAFe Agilist, Scaled Agile

Professional Scrum Master II, Scrum.org

Professional Scrum Product Owner I, Scrum.org

INDEPENDENT PRODUCTS

The Dry Standard – Production non-alcoholic drink catalog & comparison engine.

OPEN SOURCE

bb-run – Run Bitbucket Pipelines locally.

testrisk – Rank the highest-value Python test gaps from coverage, AST, and git.

TECHNICAL EXPERTISE

Languages: Python, TypeScript, Java, Bash

Cloud & Containers: AWS, Docker, Kubernetes, Helm

CI/CD & Developer Tooling: Git, Bitbucket Pipelines, GitLab CI, GitHub Actions, release automation, security scanning

Python Tooling: uv, Ruff, ty, pytest, pre-commit – dependency management, linting, type checking, testing, and automated quality gates

Integration: REST/OpenAPI, RabbitMQ, ActiveMQ, distributed systems

Data: PostgreSQL, MySQL, SQL Server, Elasticsearch/OpenSearch, Redis